Capture the Flag 3: Data-Driven UI

Goal: Separate application data from the UI and handle multiple pieces of data in a usable interface.

Instructor Note

These activities require your own problem-solving and research skills.

DO NOT use AI to generate the code for you.

You may use AI to:

- Research what a Dart/Flutter feature does
- Understand syntax
- **Explain** an error (not **fix** your error)
- Clarify documentation

You may **NOT** ask AI to generate the solution/code for a flag.

🏴 Flag 0 — Create Your Feature Branch

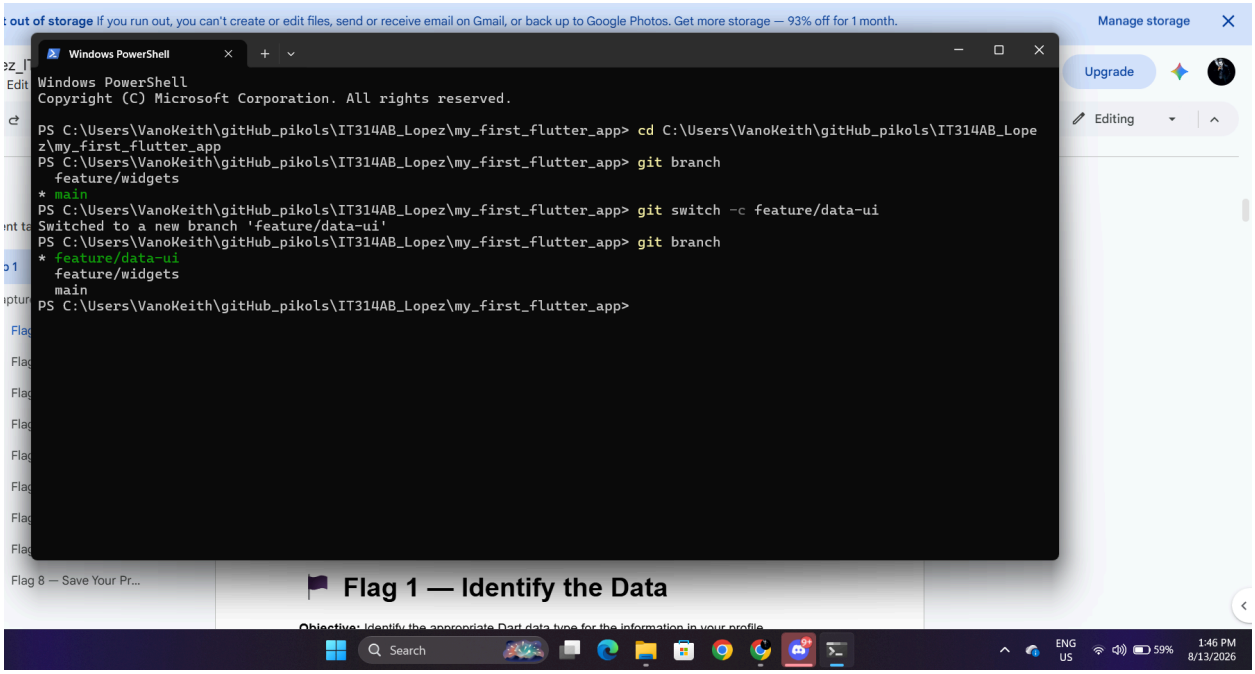
Objective: Create a feature branch before modifying your application.

Create:

`feature/data-ui`

Screenshot:

- Terminal showing your current branch



🏴 Flag 1 — Identify the Data

Objective: Identify the appropriate Dart data type for the information in your profile.

Research basic Dart data types and identify at least **four different types** that could be used in your application.

Examples:

String
int
double
bool

Create a table identifying the information and its appropriate data type.

Create:

- Your identified data
- The corresponding data types

name	**string**
age	**int**
height	**double**
student	**bool**

🏴 Flag 2 — Create Your Data

Objective: Create appropriately typed variables for your profile.

Create variables for at least:

- Name
- Course & Section
- Age
- Hobby
- Height
- Student status

Your data should exist **separately from your UI widgets**.

Screenshot:

- Your variables

```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  final String name = 'Keith Francis G. Lopez';
  final String courseSection = 'BSIT-3';
  final int age = 21;
  final String hobby = 'Touching Grass';
  final double height = 170.0;
  final bool isStudent = true;

  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'My First Flutter Application',

      home: Scaffold(
        backgroundColor: Colors.lightBlue[50]
```

- Their data types

```dart
            Text(
              'Birthdate: September 25, 2004',
              style: TextStyle(fontSize: 15),
            ), Text
          ],
        ), Row
        const SizedBox(height: 15),

        Text(
          'Height: $height cm',
          style: const TextStyle(fontSize: 15),
        ), Text

        const SizedBox(height: 10),

        Text(
          'Student: $isStudent',
          style: const TextStyle(fontSize: 15),
        ), Text
      ],
    ), Column
  ), Padding
), Card
```

- Application still running



🏴 Flag 3 — Connect Data to UI

Objective: Replace your hardcoded information with your variables.

Your profile should now display information using the variables you created.

Test:

Change the value of one of your variables.

The UI should automatically display the new value.

Screenshot:

- Variables

- Widgets displaying the variables



```
74          Text(
75            name,
76            textAlign: TextAlign.center,
77            style: TextStyle(
78              fontSize: 28,
79              fontWeight: FontWeight.bold,
80              color: Colors.blue,
81            ), TextStyle
82          ), Text
83
84          const SizedBox(height: 15),
85
86          Text(
87            courseSection,
88            style: TextStyle(
89              fontSize: 22,
90              fontWeight: FontWeight.bold,
91            ), TextStyle
92          ), Text
93
```

```
117          Text(
118            'Hobby: $hobby',
119            style: const TextStyle(
120              fontSize: 18,
121            ), TextStyle
122          ), Text
123
124          const SizedBox(height: 15),
125
126          const Divider(),
127
128          const SizedBox(height: 15),
129
130          Row(
131            mainAxisAlignment: MainAxisAlignment.center,
132            crossAxisAlignment: CrossAxisAlignment.center,
133            children: [
134              Text(
135                'Age: $age',
136                style: const TextStyle(fontSize: 15),
137              ), Text
138
139              SizedBox(width: 30),
140
```

- Updated application



🏴 Flag 4 — Make the Image Data-Driven

Objective: Make your profile image part of your data.

Create a variable containing the path to your profile image.

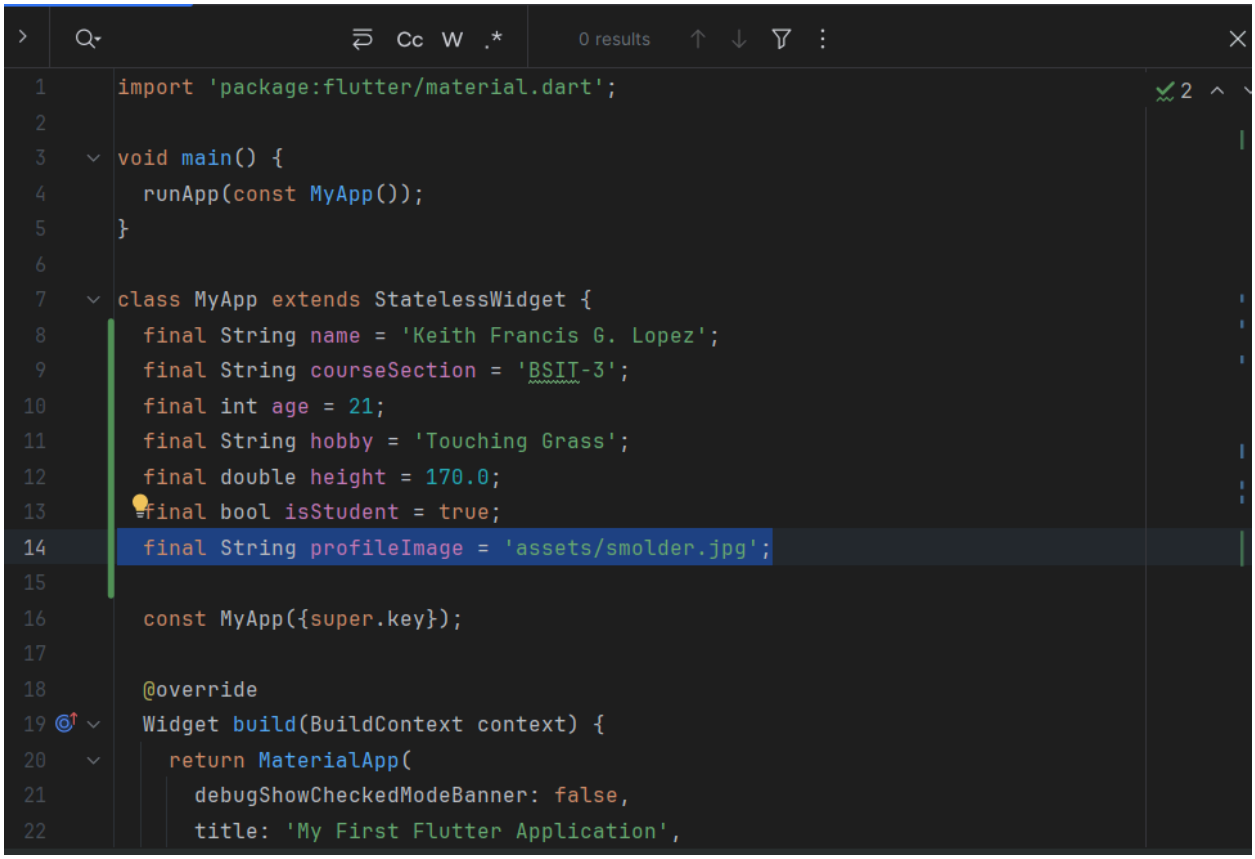
Use the variable when displaying the image.

Then change the variable to another registered image.

The image displayed by the application should change.

Screenshot:

- Image variable



```dart
import 'package:flutter/material.dart';

void main() {
  runApp(const MyApp());
}

class MyApp extends StatelessWidget {
  final String name = 'Keith Francis G. Lopez';
  final String courseSection = 'BSIT-3';
  final int age = 21;
  final String hobby = 'Touching Grass';
  final double height = 170.0;
  final bool isStudent = true;
  final String profileImage = 'assets/smolder.jpg';

  const MyApp({super.key});

  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      title: 'My First Flutter Application',
```

- Image widget using the variable



```dart
            child: Column(
              mainAxisSize: MainAxisSize.min,
              children: [
                ClipRRect(
                  borderRadius: BorderRadius.circular(20),
                  child: Image.asset(
                    profileImage,
                    width: 200,
                    height: 200,
                    fit: BoxFit.cover,
                  ),
                ),
```

- Updated application



🏴 Flag 5 — Five Profiles, One Design

Objective: Create at least **five different profile cards** using the same visual design.

Each profile must contain at least:

- Image
- Name
- Course & Section
- Age
- Hobby

The five profiles must contain different information.

Important

All five profiles should use the **same visual design**.

You are demonstrating that the same UI can represent different data.

Screenshot:

- Your 5 different profile data

```dart
// YOUR PROFILE
ProfileCard(
  image: profileImage,
  name: name,
  courseSection: courseSection,
  age: age,
  hobby: hobby,
  birthdate: 'September 25, 2004',
), ProfileCard

const SizedBox(height: 15),

// FAVORITES
Card(
  elevation: 7,
  shape: RoundedRectangleBorder(
    borderRadius: BorderRadius.circular(18),
    side: const BorderSide(
      color: Colors.blue,
      width: 2,
    ), BorderSide
  ), RoundedRectangleBorder

  child: Padding(
    padding: const EdgeInsets.all(20),
```

```dart
138   // PROFILE 2
139   ProfileCard(
140     image: 'assets/2in1.jpg',
141     name: 'John Doe',
142     courseSection: 'BSIT-3',
143     age: 20,
144     hobby: 'Playing Online Games',
145   ), ProfileCard
146
147   const SizedBox(height: 15),
148
149   // PROFILE 3
150   ProfileCard(
151     image: 'assets/SleepingBeuty.jpg',
152     name: 'Vin Slash',
153     courseSection: 'BSIT-3',
154     age: 21,
155     hobby: 'Sleeping',
156   ), ProfileCard
157
158   const SizedBox(height: 15),
159
160   // PROFILE 4
161   ProfileCard(
162     image: 'assets/Cutie.jpg',
163     name: 'Kenwell Ivankov',
164     courseSection: 'BSIT-3',
165     age: 22,
166     hobby: 'Being Cute',
167   ), ProfileCard
168
```



```
168
169                 const SizedBox(height: 15),
170
171                 // PROFILE 5
172                 ProfileCard(
173                   image: 'assets/nasarapan.jpg',
174                   name: 'Carl Garcia',
175                   courseSection: 'BSIT-3',
176                   age: 20,
177                   hobby: 'Coding and Inlab',
178                 ), ProfileCard
179
180                 const SizedBox(height: 20),
181               ],
182             ), Column
183           ), Center
184         ), SingleChildScrollView
185       ), Scaffold
186     ); MaterialApp
```

🏴 Flag 6 — THE HIDDEN PROBLEM

You didn't see your UI?

Objective: Make all five profiles accessible to the user.

You just added five profile cards.

Now run your application.

Something is wrong.

Look carefully at your emulator.

Can you see all five profiles?

Maybe you can see the first one...

Maybe part of the second...

But where are the rest?

That's right.

You can't see them.

Your UI is larger than the available screen.

The profiles exist.

The data exists.

The widgets exist.

But the user can't access everything.

Your Mission:

Figure out how Flutter allows users to **scroll through content that is larger than the available screen**.

Research Flutter's **scrolling widgets** and determine which one is appropriate for your current layout.

You must make it possible to scroll through all five profiles.

Requirement:

The user must be able to:

Scroll → see Profile 1 → Profile 2 → Profile 3 → Profile 4 → Profile 5

without changing the size of the screen.

Screenshot:

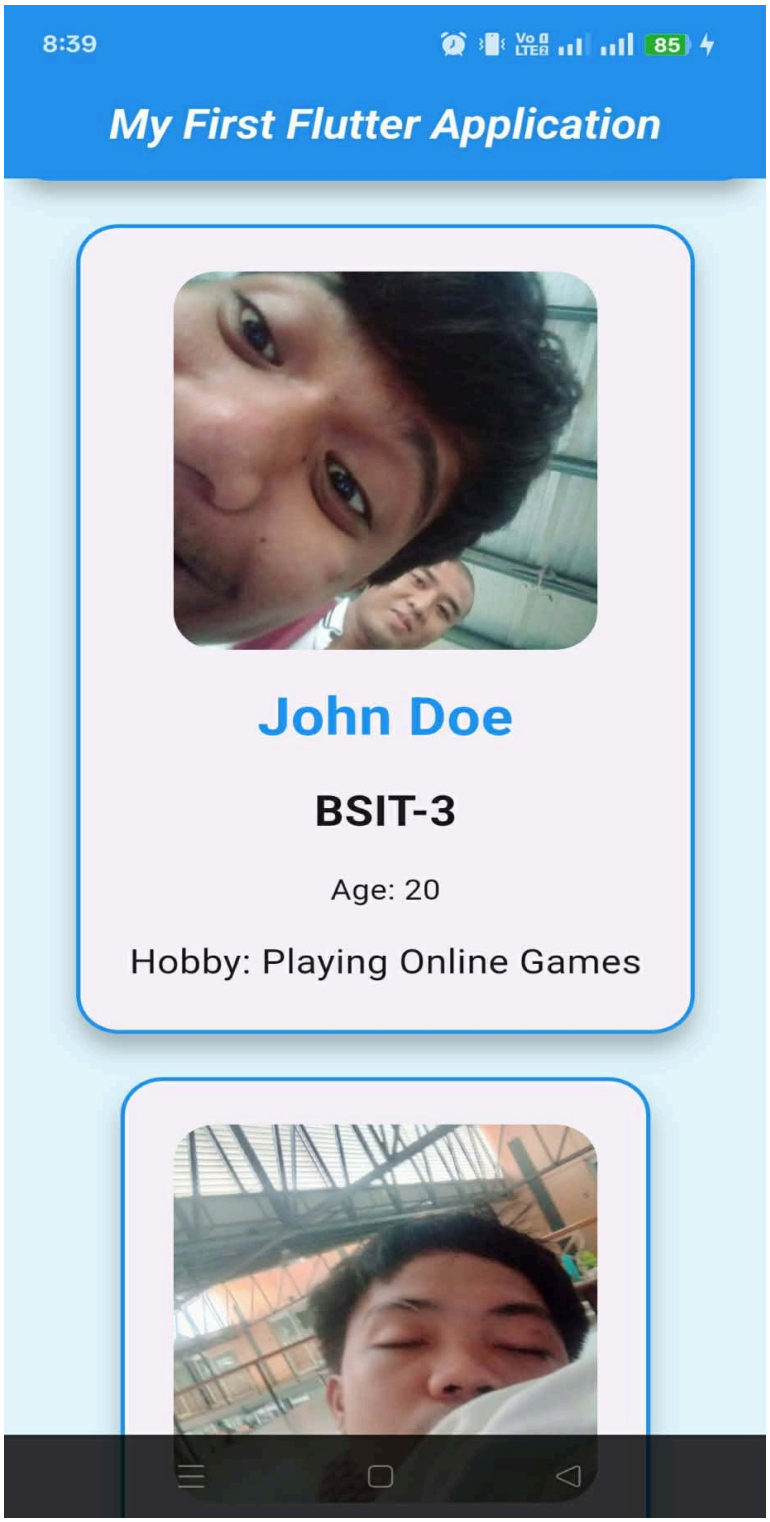
- The scrolling widget in your code

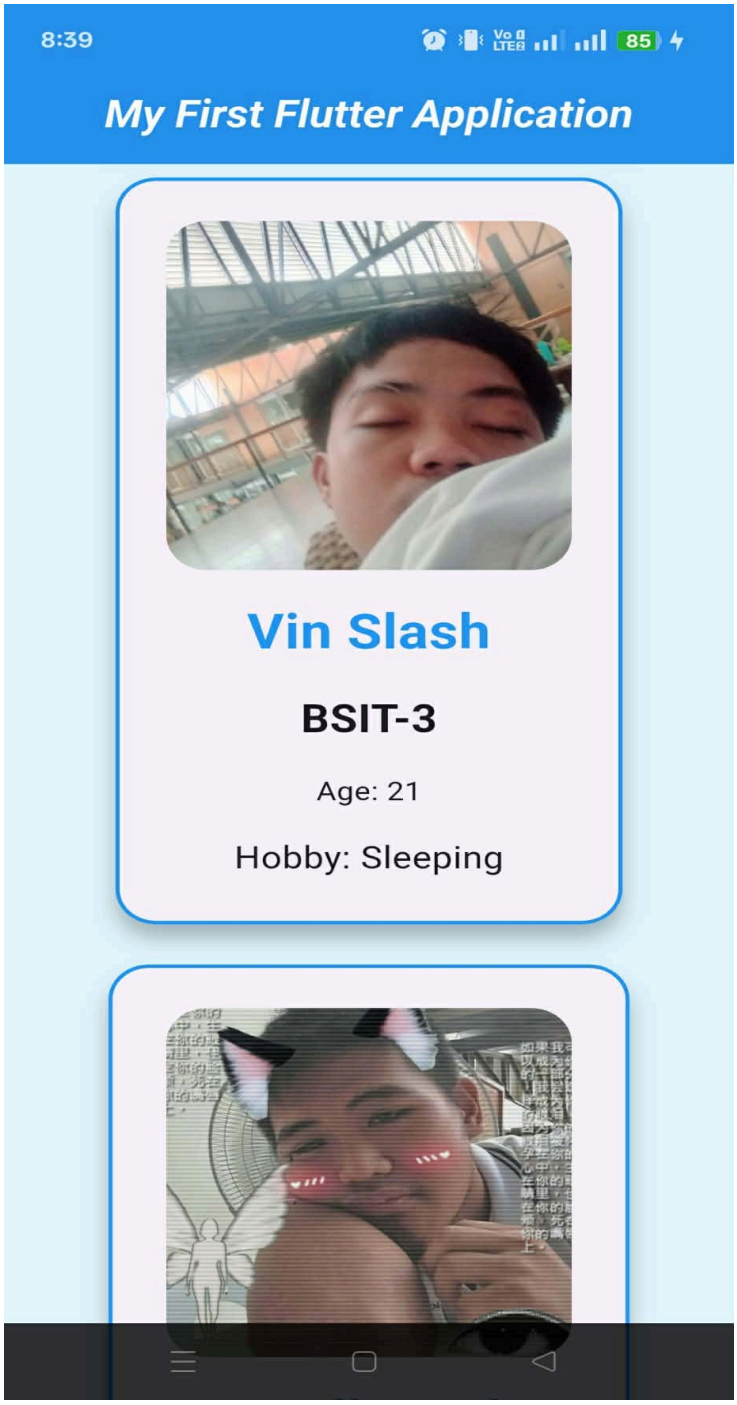
```
39
40          // ScrollView
41          body: SingleChildScrollView(
42            child: Center(
43              child: Column(
44                mainAxisAlignment: MainAxisAlignment.center,
45                children: [
46
```

- Top portion of the application



- The rest of the portion of the application showing the later profiles




8:39
My First Flutter Application
Vin Slash
BSIT-3
Age: 21
Hobby: Sleeping

8:39

My First Flutter Application

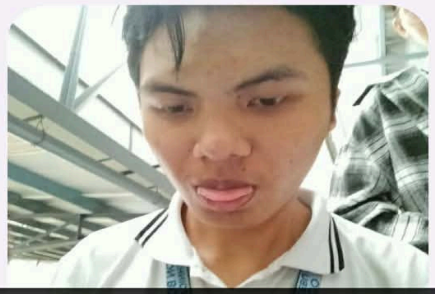


Kenwell Ivankov

BSIT-3

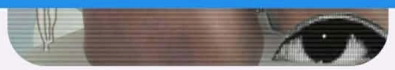
Age: 22

Hobby: Being Cute



8:39

My First Flutter Application

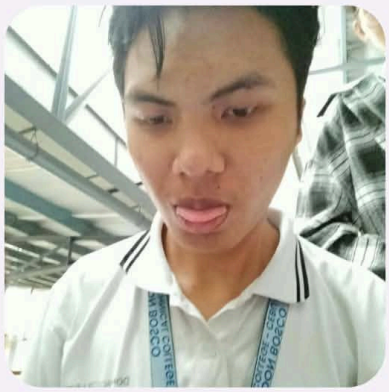


Kenwell Ivankov

BSIT-3

Age: 22

Hobby: Being Cute



Carl Garcia

BSIT-3

Age: 20

Hobby: Coding and Inlab

🏴 Flag 7 — Survive the Database

Objective: Prepare your UI for incomplete data.

Now imagine your profiles aren't manually created.

Someday, the data will come from a database.

And real data isn't always perfect.

Some information may be:

- Missing
- Empty
- `null`
- Not provided

Create at least **three profiles with missing information**.

For example:

Profile A
Name: Maria
Course: BSIT
Hobby: missing

Profile B
Name: Juan
Course: missing
Hobby: Gaming

Profile C
Name: missing
Course: MMA
Hobby: Drawing

Your application must **not crash** when information is missing.

Instead, provide an appropriate fallback.

Examples:

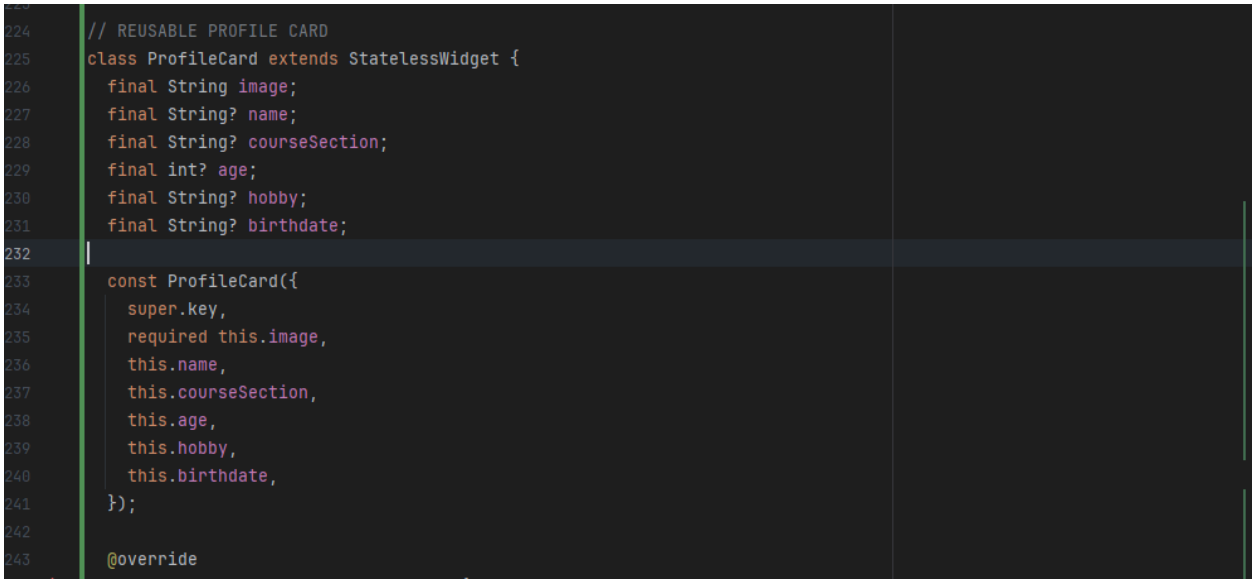
Hobby: Not provided
Course: Unknown

Challenge

Your placeholders should look intentional and clean.

Screenshot:

- Incomplete profile data

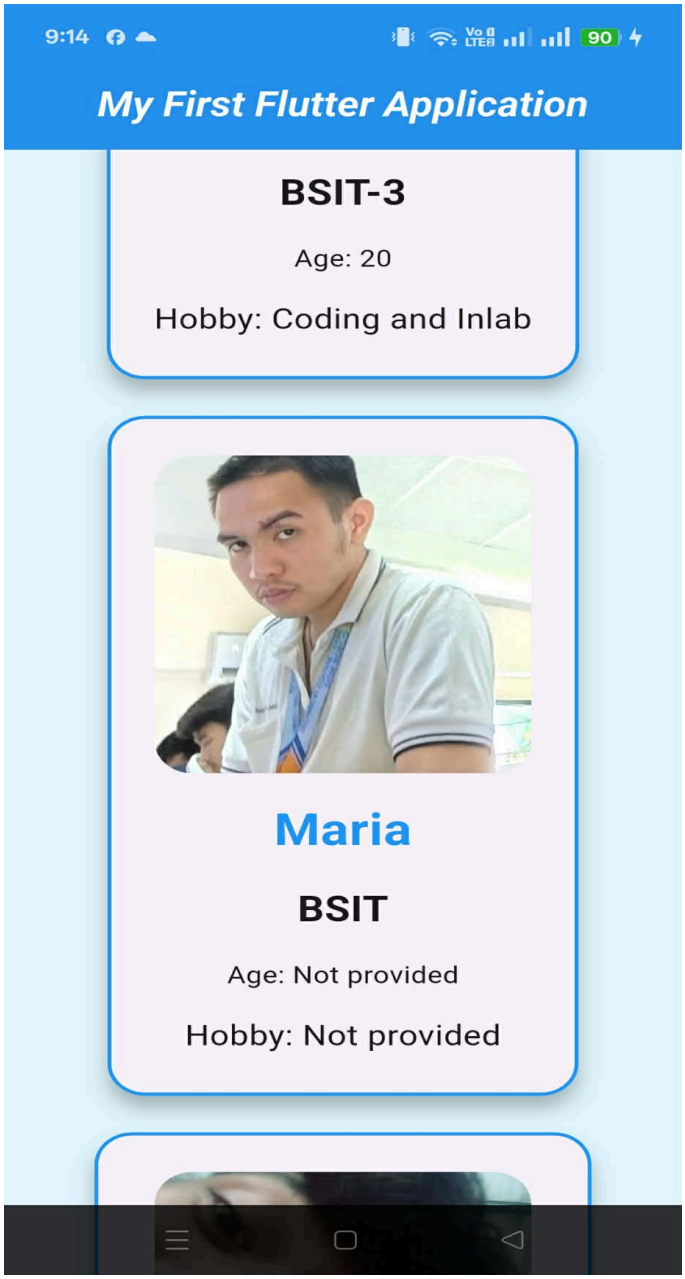


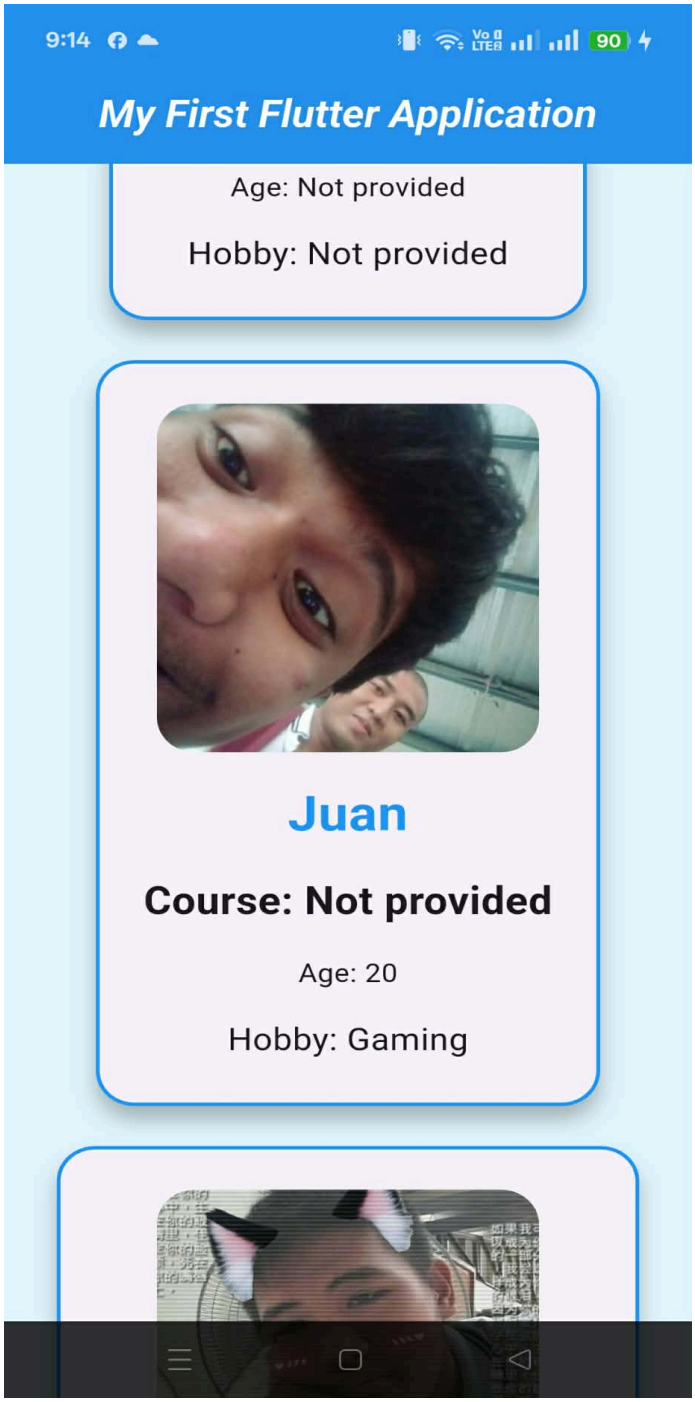
```dart
// REUSABLE PROFILE CARD
class ProfileCard extends StatelessWidget {
  final String image;
  final String? name;
  final String? courseSection;
  final int? age;
  final String? hobby;
  final String? birthdate;

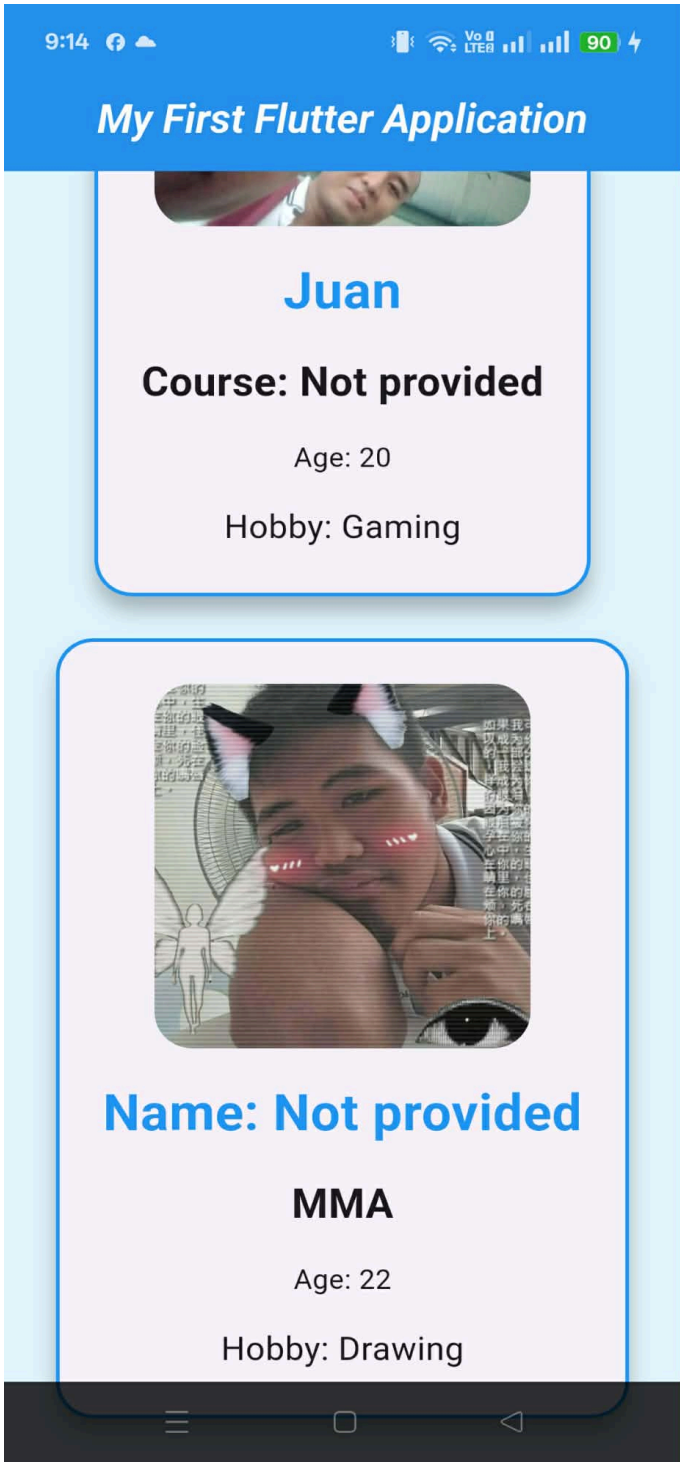
  const ProfileCard({
    super.key,
    required this.image,
    this.name,
    this.courseSection,
    this.age,
    this.hobby,
    this.birthdate,
  });

  @override
```

- Application displaying fallback values




9:14
My First Flutter Application
Age: Not provided
Hobby: Not provided
Juan
Course: Not provided
Age: 20
Hobby: Gaming


9:14
My First Flutter Application
Juan
Course: Not provided
Age: 20
Hobby: Gaming
Name: Not provided
MMA
Age: 22
Hobby: Drawing

- Application still functioning correctly



8:39

My First Flutter Application

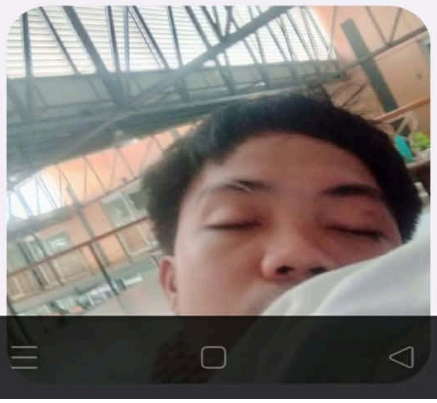


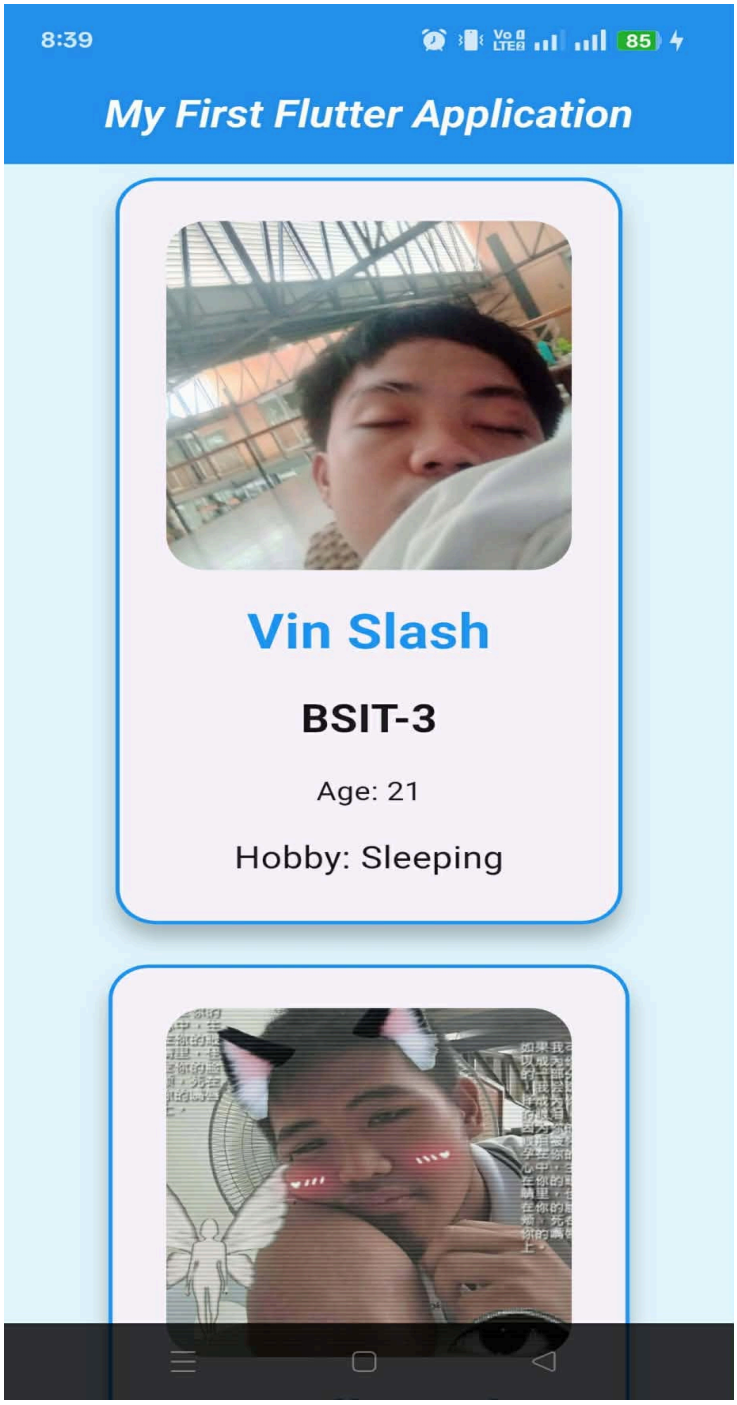
John Doe

BSIT-3

Age: 20

Hobby: Playing Online Games




8:39
My First Flutter Application
Vin Slash
BSIT-3
Age: 21
Hobby: Sleeping

8:39

My First Flutter Application

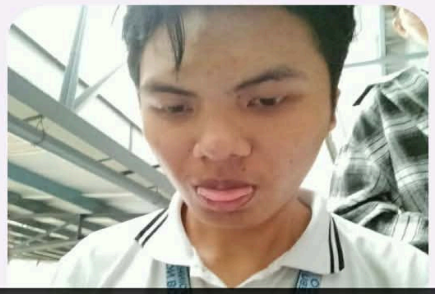


Kenwell Ivankov

BSIT-3

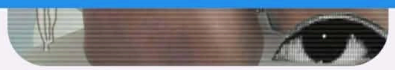
Age: 22

Hobby: Being Cute



8:39

My First Flutter Application

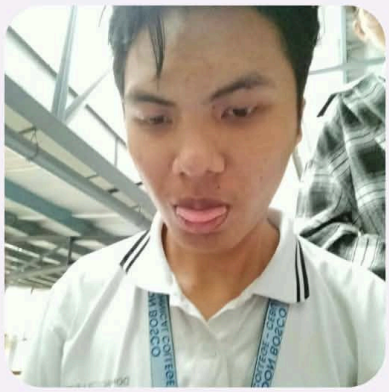


Kenwell Ivankov

BSIT-3

Age: 22

Hobby: Being Cute



Carl Garcia

BSIT-3

Age: 20

Hobby: Coding and Inlab

🏴 Flag 8 — Save Your Progress

Objective: Commit and push your completed Data-Driven UI CTF.

Use an appropriate commit message following proper Git standards.

Branch:

`feature/data-ui`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

